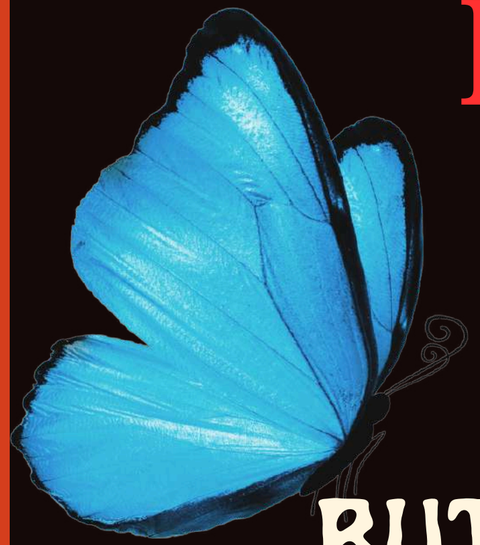

ELEMENTS OF BIOLOGY

Presentation

BUTTERFLY WING NANOSTRUCTURES

BIOMIMICRY

PPT Presented by

Aaryan Mishra(I045), Satvik Shankar(I064), Poorvi Soni(I07I), Anirban Dey(I083)

AGENDAS

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02 WHAT ARE BLUE MORPHO BUTTERFLIES?

03 BUTTERFLY WING NANOSTRUCTURE

04 ROLE OF IRIDESCENCE AND INTERFERENCE

05 IMPLEMENTATION

06 APPLICATIONS IN BANKING NOTES

07 APPLICATIONS IN HOLOGRAMS

08 APPLICATIONS IN SOLAR CELLS

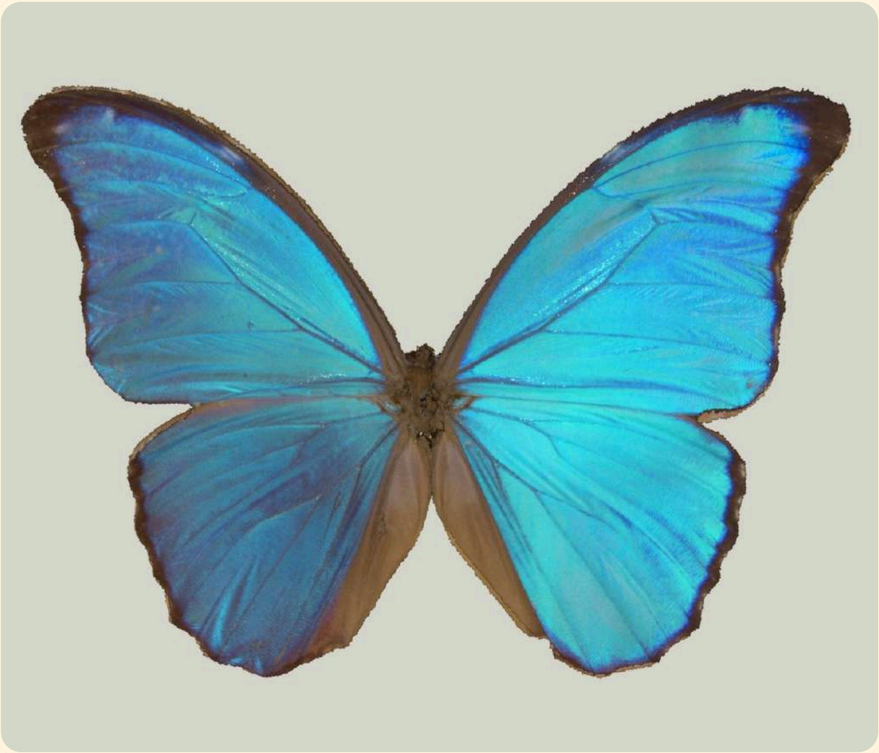
09 PROS AND CONS

10 STATISTICAL DATA

11 RESEARCH TIMELINE

12 CONCLUSION

These striking insects are found in the tropical forests of Central and South America. With a wingspan of about 5 to 8 inches, they are among the largest butterflies in the world. Their brilliant, metallic blue color doesn't come from pigments but from tiny micro- and nano-scale structures on their wings that reflect and manipulate light.



The upper side of their wings shines with a bright, shimmering blue, while the underside is a softer brown. These lower surfaces also have eye-like spots that help protect them by blending in with their surroundings when the wings are closed. Butterfly wings are made up of thousands of tiny overlapping scales that provide both insulation and color.

WHAT IS BLUE MORPHO BUTTERFLY?

https://www.rainforest-alliance.org/species/blue-morpho-butterfly/

Counterfeiting around the world is expected to grow to nearly $4.2 trillion, putting both the global economy and public safety at risk. With the help of modern high-resolution technology like digital cameras, scanners, and printers, criminals can now produce very realistic fake products more easily than before.

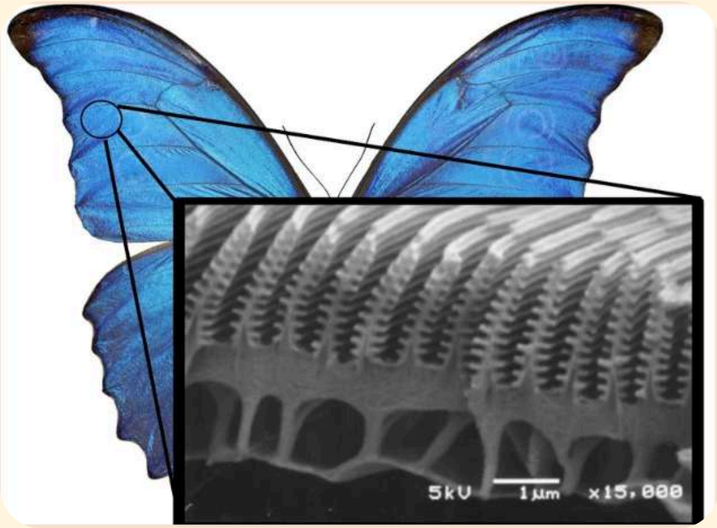
PROBLEM STATEMENT

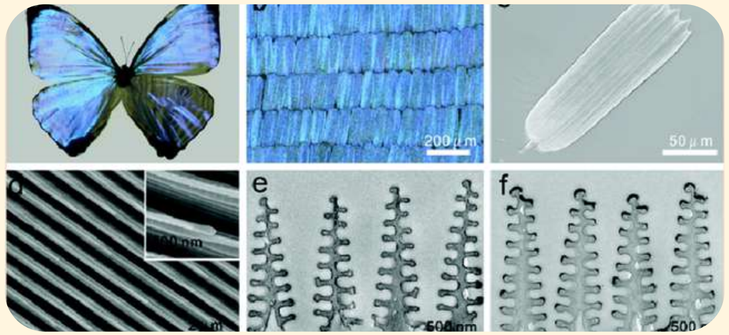
Conventional security features are often easy to replicate because they rely on patterns that can be copied using common industrial equipment that is widely available.

https://alpvision.com/proven-anti-counterfeiting-measures-for-your-brand/

https://sloanreview.mit.edu/article/taming-the-counterfeiting-epidemic/

BUTTERFLY WINGS NANOSTRUCTURE





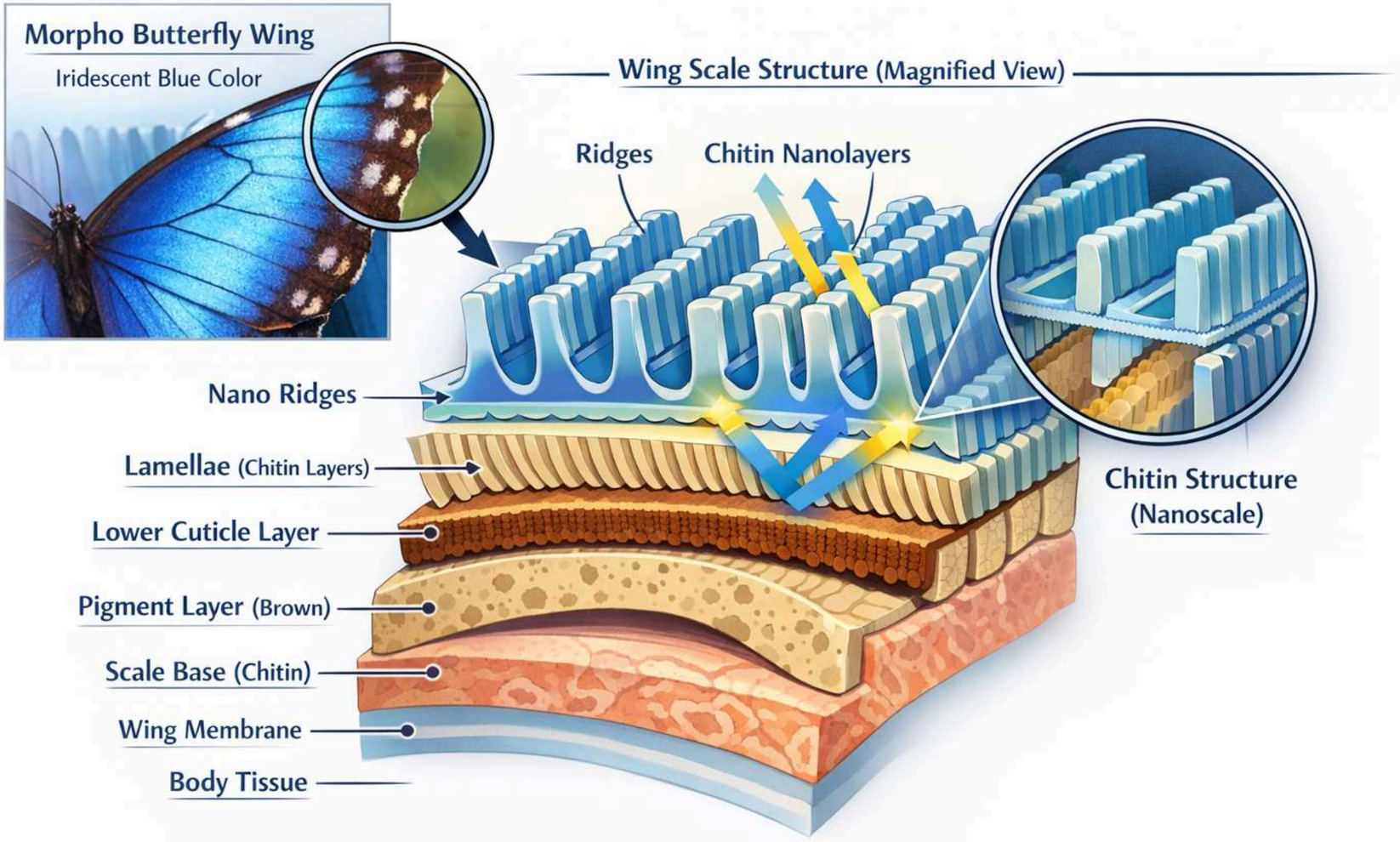
- Each wing is covered with over a million overlapping scales, each
- about 70-100 µm long and 1-2 µm thick. Each scale is made of
- chitin, a transparent, lightweight, and tough biopolymer.

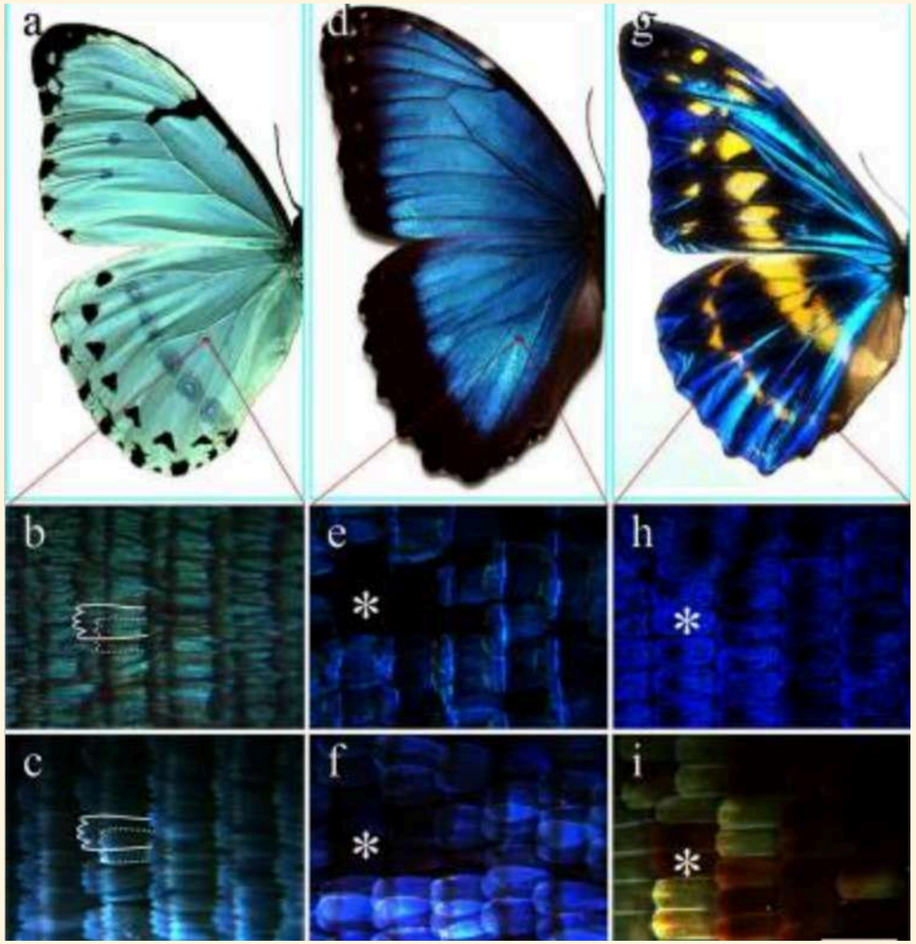
- Each scale has two parts:
  - Lower lamina (base plate): A flat, smooth surface that reflects a small. amount of light.
  - Upper lamina (ridge layer): A set of parallel ridges running along the scale, each ridge being a stack of thin, horizontal lamellae resembling a Christmas tree in cross-section.
  - Between these ridges are microscopic tree-like branches (called lamellae) which are spaced ~200-300 nm apart, resulting in the creation of a multi-layered air-chitin structure

- This coordinated combination of nanostructures (in the range of 1 to 100 nm) helps in manipulating light through optical interference, transforming it into the dazzling electric blue colour we observe.

https://www.sciencedirect.com/science/article/abs/pii/S0959437X21000058
https://www.uvm.edu/~dahammon/Structural_Colors/Structural_Colors/Structual_Colors_In_Butterflies.html

Morpho Butterfly Wing Nanostructure




a
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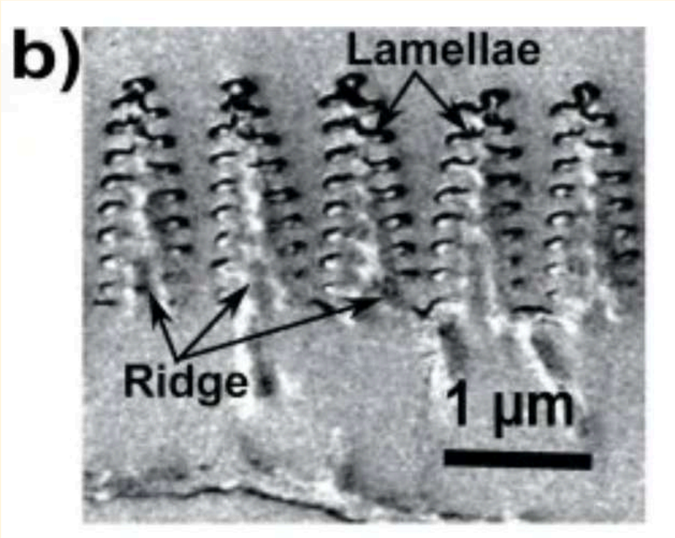
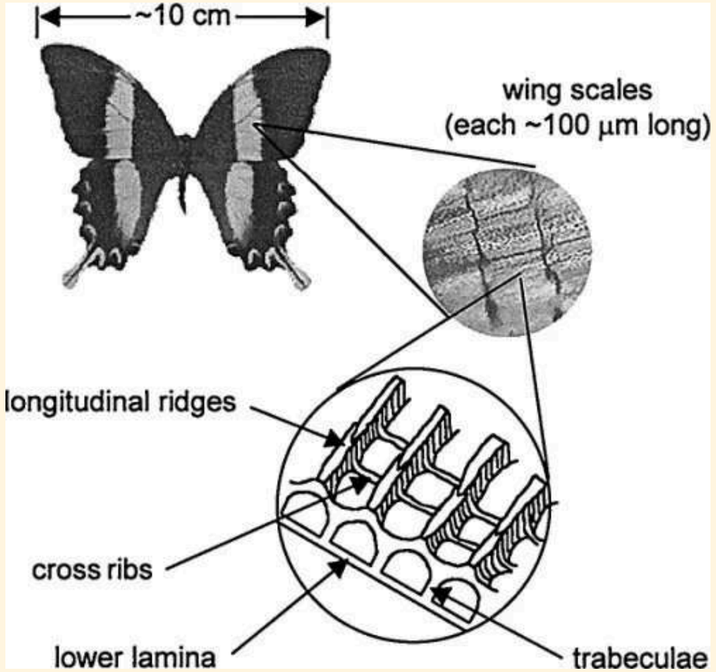
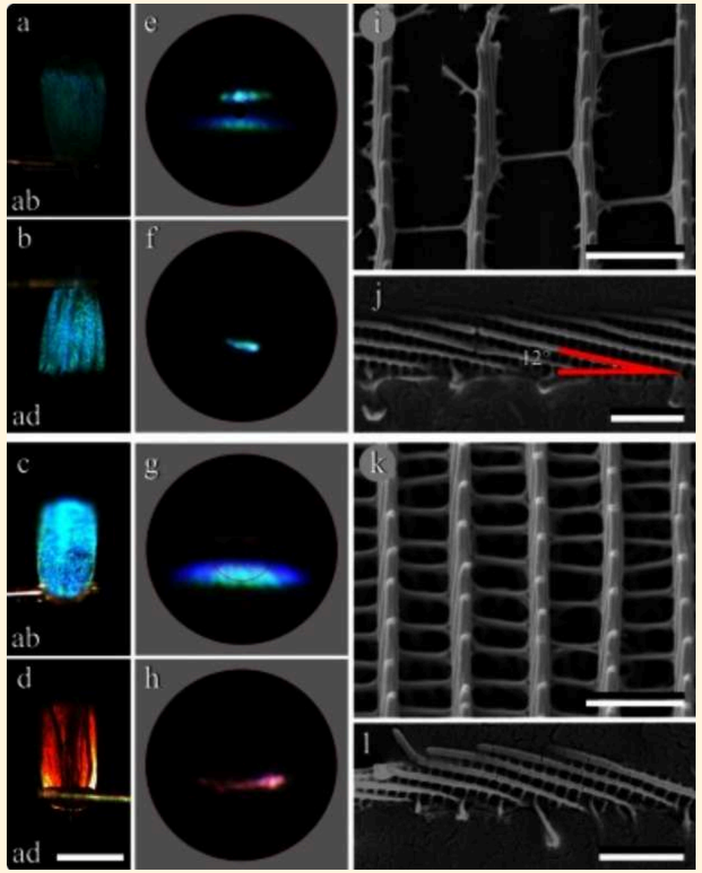

b)
Lamellae
Ridge
1 µm


~10 cm
wing scales
(each ~100 µm long)
longitudinal ridges
cross ribs
lower lamina
trabeculae


a
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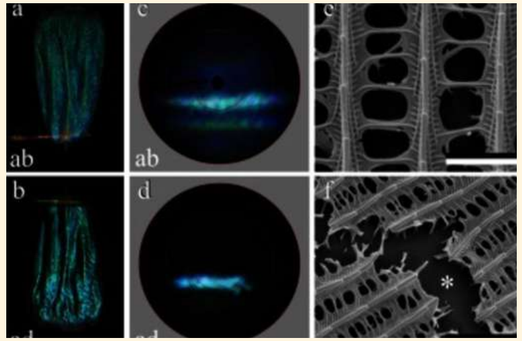
ROLE OF IRIDESCENCE AND OPTICAL INTERFERENCE

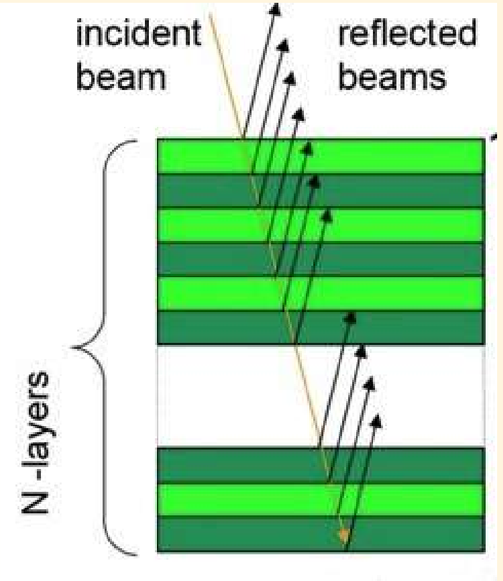
<u>Thin-Film Interference</u>

The lower lamina of the wing scale works like a thin-film reflector, while the upper multilayered lamina enhances the brightness of the reflection. The ridges, made of alternating layers of chitin and air with specific spacing, produce the structural colour effect.

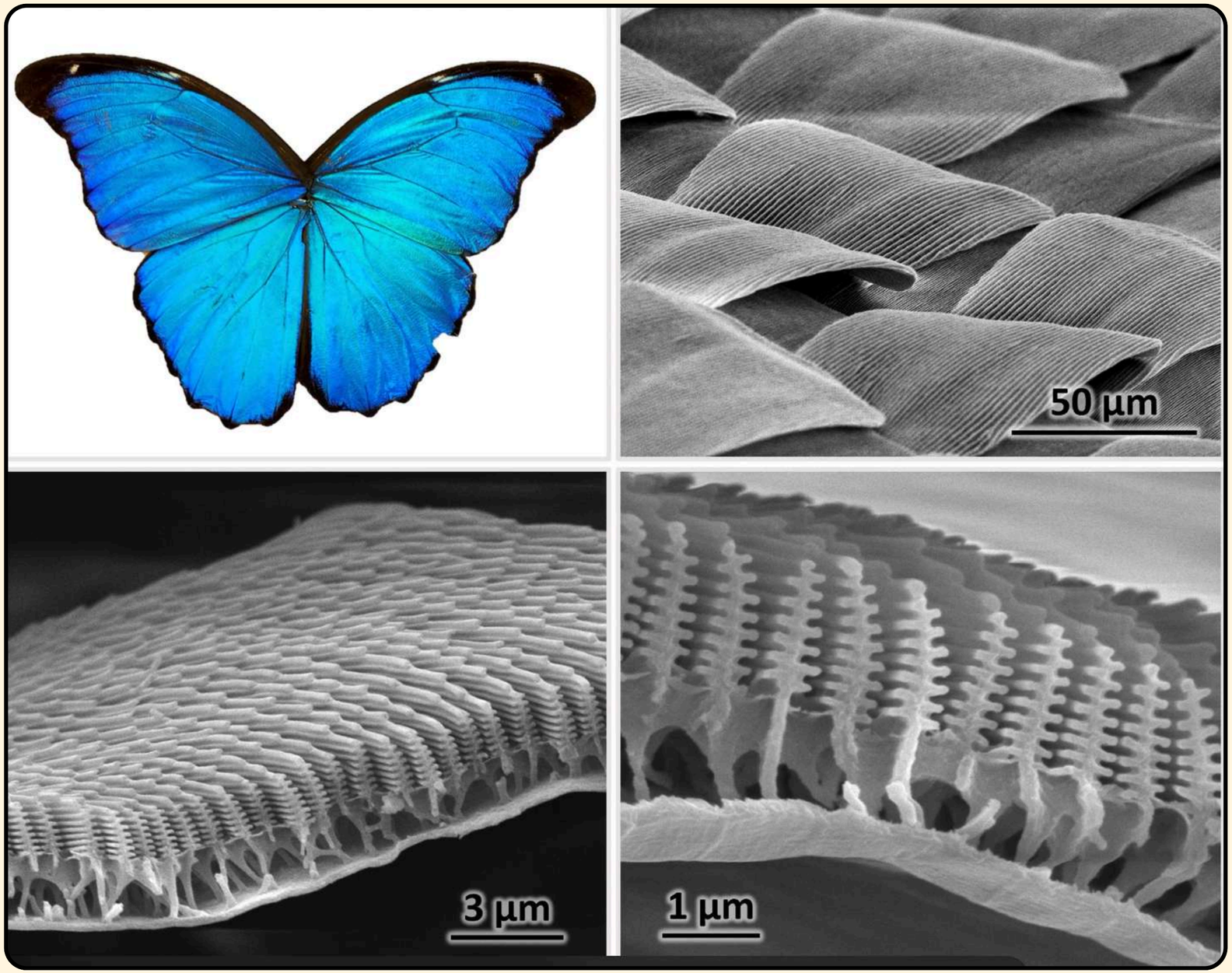
These tree-like ridges are arranged as layers of materials with different refractive indices. When white light hits the scale,

The reflected waves interact with each other. When the path difference matches a multiple of the blue wavelength,constructive interference occurs., making blue light appear stronger. Other wavelengths such as red or green undergo destructive interference, so they cancel out and are not seen.



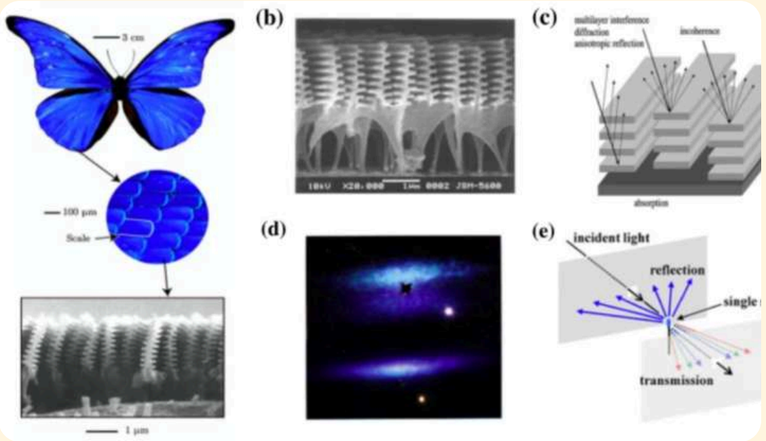


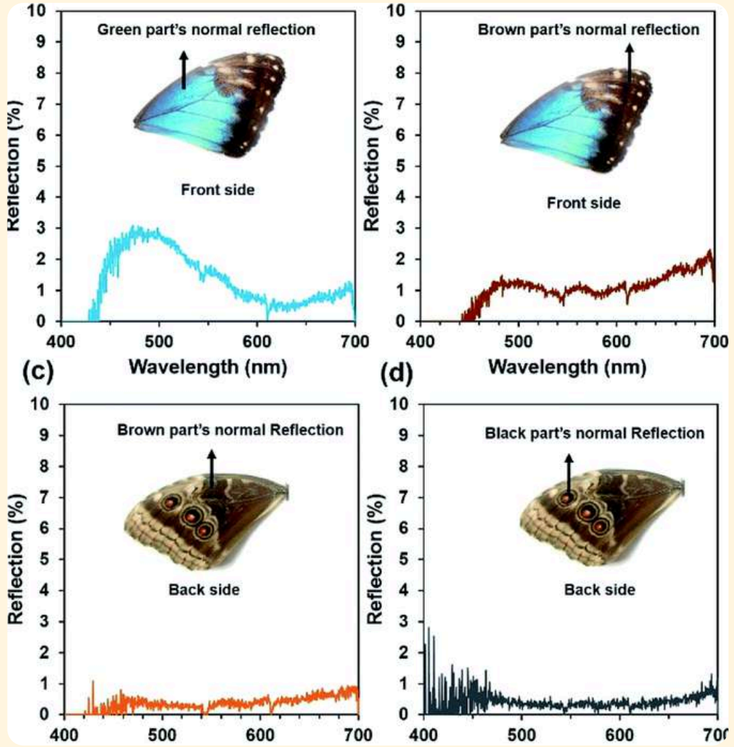
https://pmc.ncbi.nlm.nih.gov/articles/PMC4841846/
https://journals.biologists.com/jeb/article/219/24/3936/16551/Coloration-mechanisms-and-phylogeny-of-Morpho


50 µm
3 µm
1 µm

DIFFRACTION

The ridges on the wing scales are spaced almost the same as the wavelength of light. Because of this, they behave like a diffraction grating, splitting and redirecting incoming light in different directions. This process increases the intensity of the bright blue shimmer seen on the wings.





IRIDESCENCE

Iridescence is when a surface appears to change colour depending on the viewing or lighting angle.

On the butterfly's wings, the reflected light waves overlap and interfere with each other—sometimes strengthening the light and sometimes cancelling it out.

Since the path difference between the waves changes with the angle of light, different wavelengths meet the interference condition at different angles.

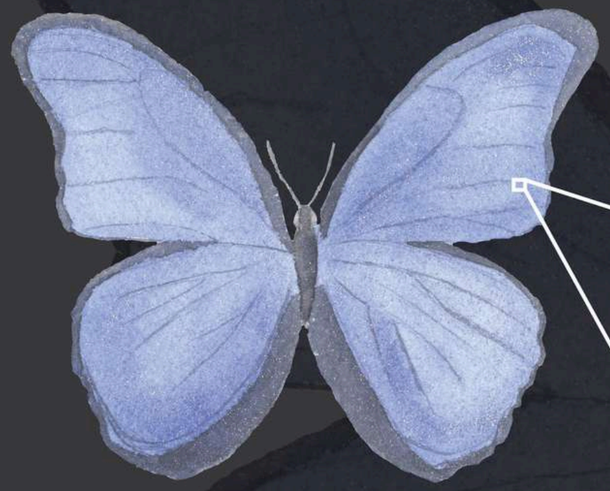
Because of the arrangement of the ridges and lamellae, light is reflected in slightly different directions, blending together to create the characteristic bright blue flash on the butterfly's wingsp.

https://www.uvm.edu/~dahammon/Structural_Colors/Structural_Colors/The_Blue_Morpho_Butterfly.html#:~:text=The%20blue%20structural%20color%20is,blue%20color%20with%20viewing%20angle

STRUCTURAL COLOR

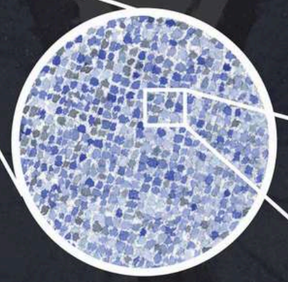
When light transforms into art

What gives the blue morpho butterfly such particular hue if no blue pigment is involved?

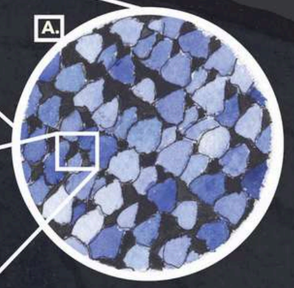


Butterfly wings are composed of scales.
Each scale constitutes one cell.
Their dimensions are at least 15 times bigger than the average human red blood cell!

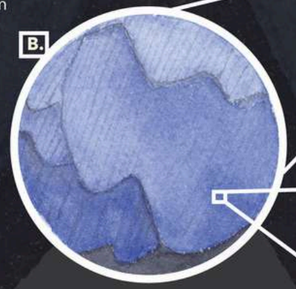
Butterfly scale



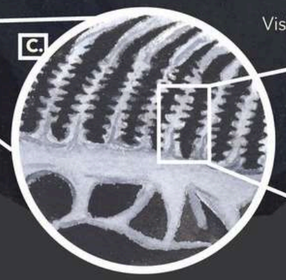
A. Scales are organized in overlapping rows, like tiles on a roof.



B. Each scale is composed of thousands of ridges.



D. Multilayer interference occurs in each layer of the pine tree shaped structure reinforcing the wavelength of blue light that our eyes perceive.



C. Ridges in each scale are composed of nanostructures (pine tree shaped) that confer the blue color.

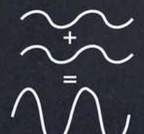
Structural color comes strictly from a purely physical origin:

INTERFERENCE OF LIGHT

When light hits a surface and bounces off to our eyes, light waves reflecting from the upper or lower side of the surface will interfere with each other.

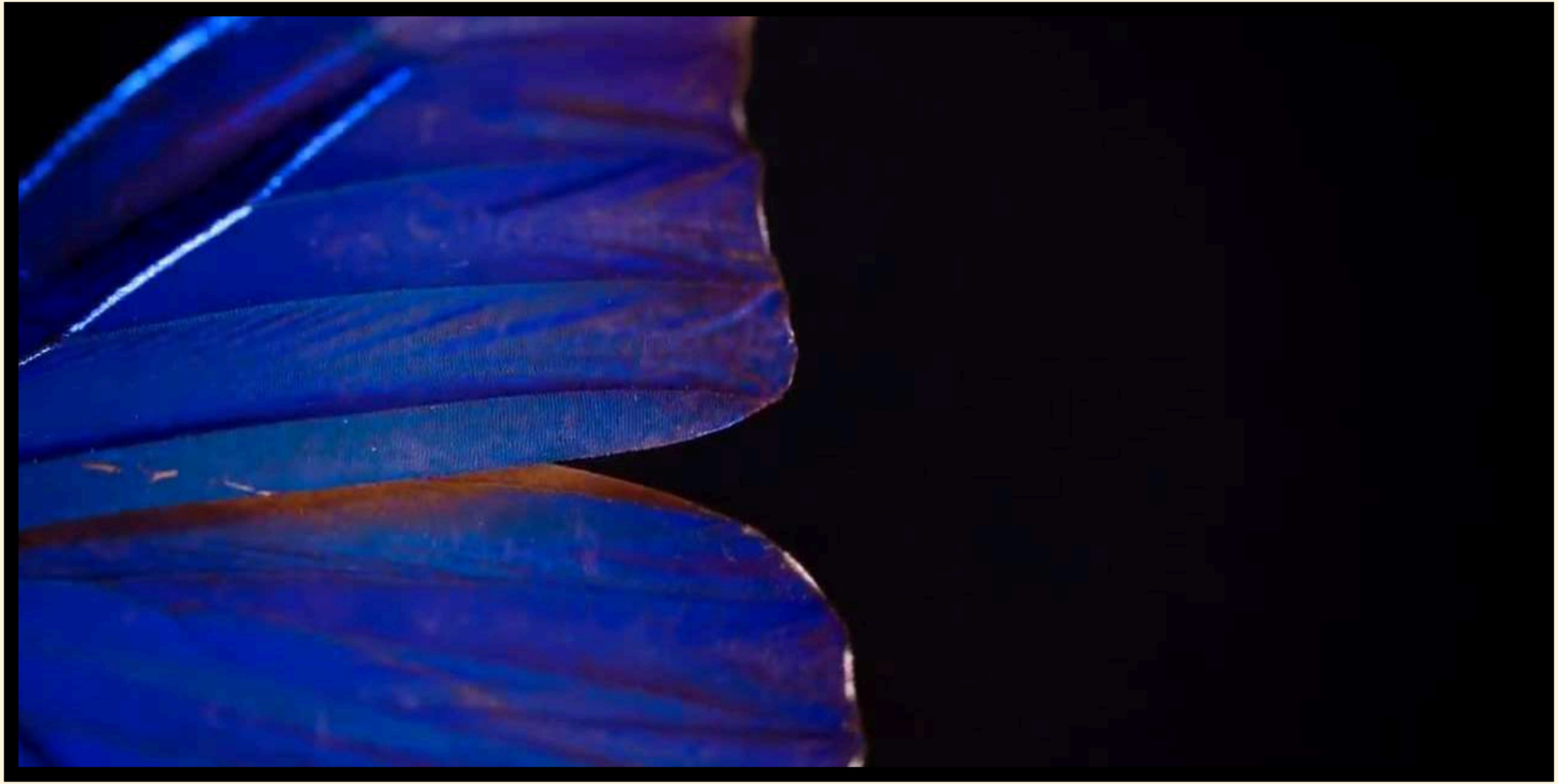
When two light waves add to form a bigger wave, this is known as:

CONSTRUCTIVE INTERFERENCE



Nanostructures in the upper side of the butterfly wing are responsible for constructive interference specfically to reinforce blue light.

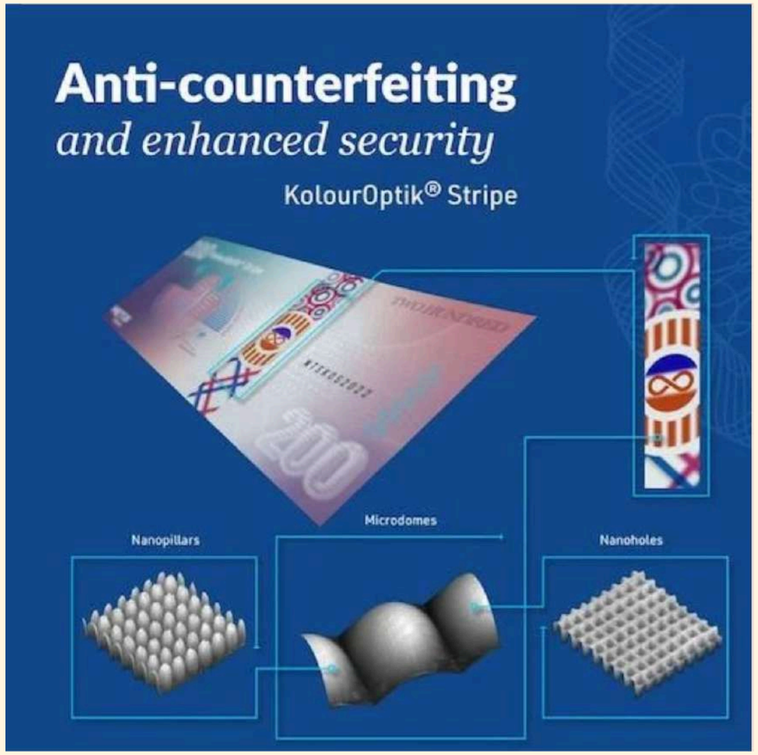
The underside of the wing is colored by brown, yellow and black pigments. The pigment absorbs unnecesary green to red light enhancing the contrast of blue color.



IMPLEMENTATION OF NANOSTRUCTURE

KolourOptik

KolourOptik combines nanoscale and microscale structures to create ultra-thin, full-colour optical security features without using traditional pigments.



I. Nanostructures in KolourOptik Strips

Designing the Nanostructures

Engineers create tiny patterns like sub-wavelength gratings and nano-pillars spaced about 100-300 nm apart, similar to visible light wavelengths. These structures are designed to reflect specific colours through plasmonic resonance and interference.

Plasmonic Colour Pixels

Instead of pigments, KolourOptik uses plasmonic pixels—metal-dielectric nanostructures that control electron movement when light hits them. Each pixel is about one-tenth the width of a human hair, and changing their size or spacing produces different colours.

Ultra-Thin Microstructure Integration

These nanostructures are embedded in microstructures under 5 micrometres thick, usually made from high-refractive-index polymers, forming thin security stripes or threads used for authentication

https://metamaterial.com/meta-materials-launches-worlds-first-fully-animated-nano-optic-banknote-security-product/ https://metamaterial.com/files/wp-content/uploads/2022/09/kolouroptikstripe_brochure_digital.pdf https://metamaterial.com/files/hubfs/1871983/20230911_quantum_brochure_digital.pdf

2. The Science at Work – Light-Structure Interaction

KolourOptik uses plasmonic nanostructures that interact with light through two main mechanisms:

MECHANISM	DESCRIPTION	RESULT
Localized Surface Plasmon Resonance (LSPR)	Collective oscillation of electrons on a metallic nanostructure's surface when illuminated	Produces vivid, tunable structural colours without pigments
Diffraction & Interference	Light waves interact with periodic nanostructure patterns	Enables angle-dependent colour shifts and dynamic effects

Together, these effects generate ultra-bright, high-contrast, non-reproducible colours that change dynamically with movement impossible to print or scan accurately.

3. Commercial Development & Manufacturing

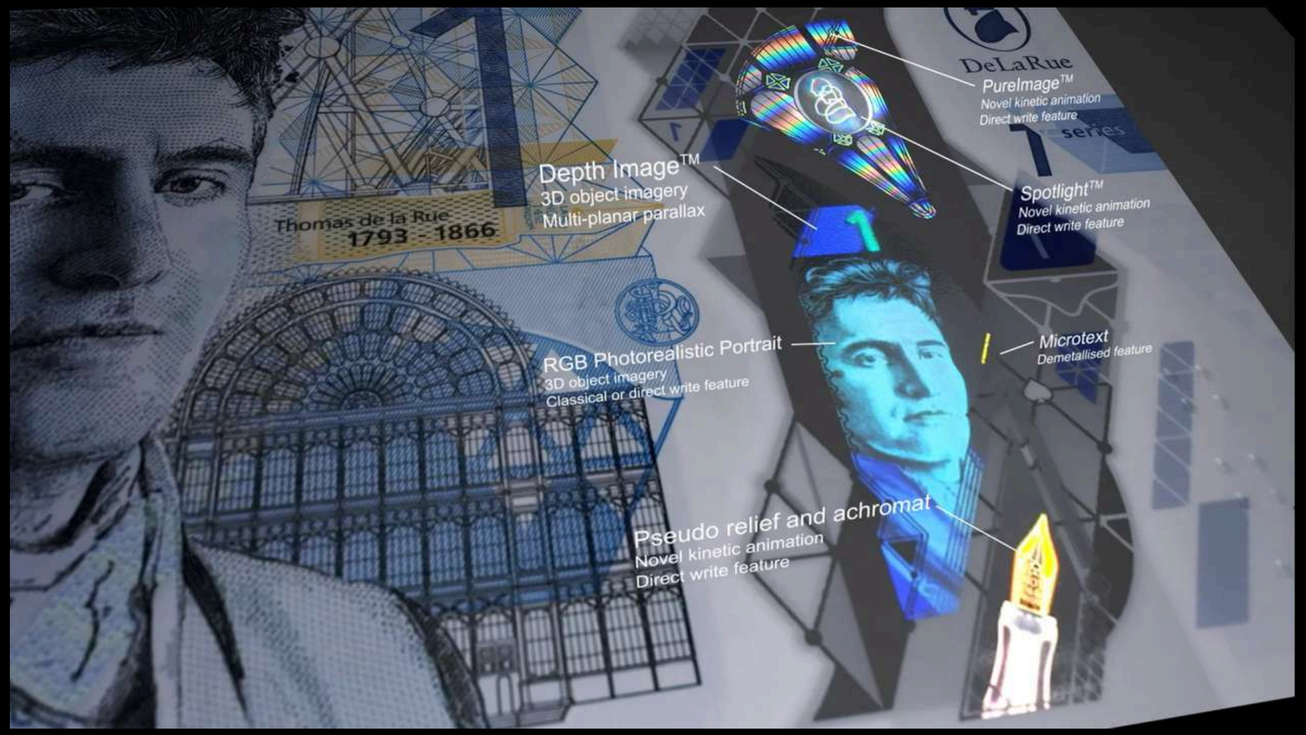
1. Backed by over 75 patents and dual 100 kV electron-beam lithography processes.
2. Manufacturing cost & complexity create a strong anti-counterfeiting barrier:
    a. 100 kV electron beam lithography machine ≈ USD 4 million each
    b. Class 100 cleanroom setup ≈ USD 50 million + USD 2 million/year maintenance
3. Production requires high-precision nanofabrication equipment, unavailable to counterfeiters.

APPLICATION IN BANK NOTES

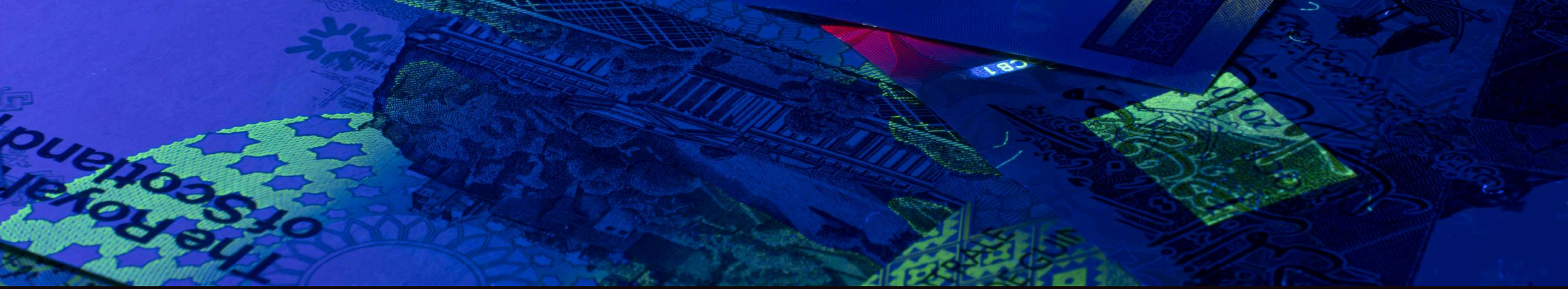
- Banknotes require extremely advanced security features as they are widely used and produced in large numbers becoming High-value targets for counterfeiters
- As counterfeiters develop more sophisticated methods, security technology has shifted toward nanoscale optical features.
- Systems such as KolourOptik demonstrate how nano-optic structures can create complex visual effects that are very difficult to copy.
- One challenge:- is ensuring these delicate nanostructures remain firmly attached and resistant to wear on the flexible materials used in currency.

- Engineering Solution: Embed nanostructures within rigid polymer or hybrid cores, ensuring encapsulation and mechanical stability—maintaining nanoscale precision and long-term durability in high-stress currency circulation.
- Unlike simple colour-shift designs, nano-optic platforms can produce vivid colours, 3D parallax effects, and even moving visual patterns.
- these effects require precise control of light at both the macro and nanoscale, they cannot be replicated using conventional holographic methods, making counterfeiting extremely difficult.

https://www.nanotec.or.th/en/butterfly-wings-behind-anti-counterfeiting- nanotechnology/ https://www.holostik.com/the-use-of-nano-optical-images-in-security-holograms/ https://www.kinegram.com/knowledge-base/kinegram-technology/kinegram-vs-hologram


DeLaRue
PureImage™
Novel kinetic animation
Direct write feature
Depth Image™
3D object imagery
Multi-planar parallax
Spotlight™
Novel kinetic animation
Direct write feature
Thomas de la Rue
1793 - 1866
RGB Photorealistic Portrait
3D object imagery
Classical or direct write feature
Microtext
Demetallised feature
Pseudo relief and achromat
Novel kinetic animation
Direct write feature

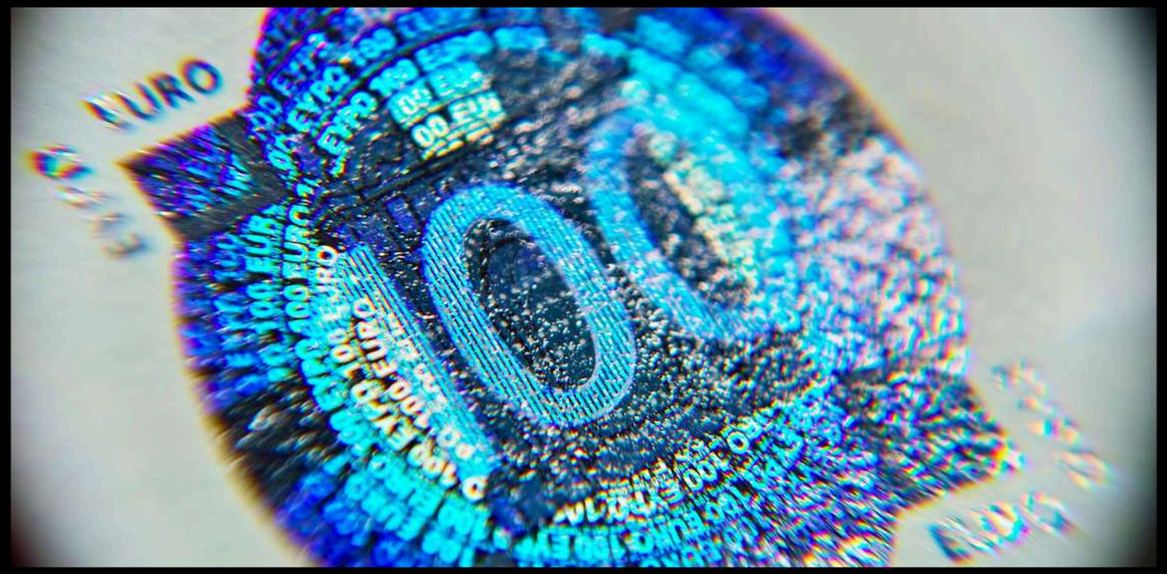
ZiG20
RESERVE BANK OF ZIMBABWE
AA0150378
Dr John Mushayavanhu
Governor
I promise to pay the bearer on demand
20
for the Reserve Bank of Zimbabwe
Harare 2024
ZiG20

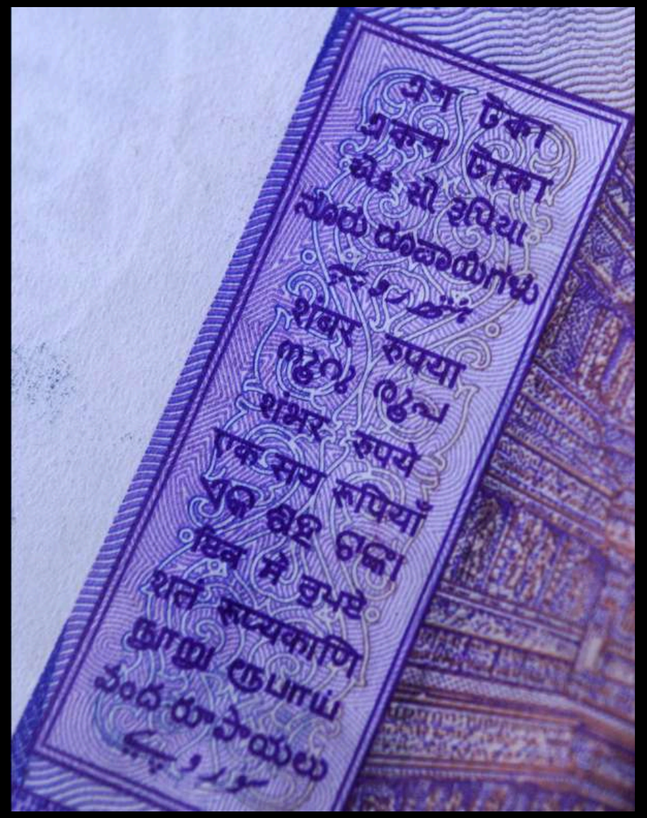


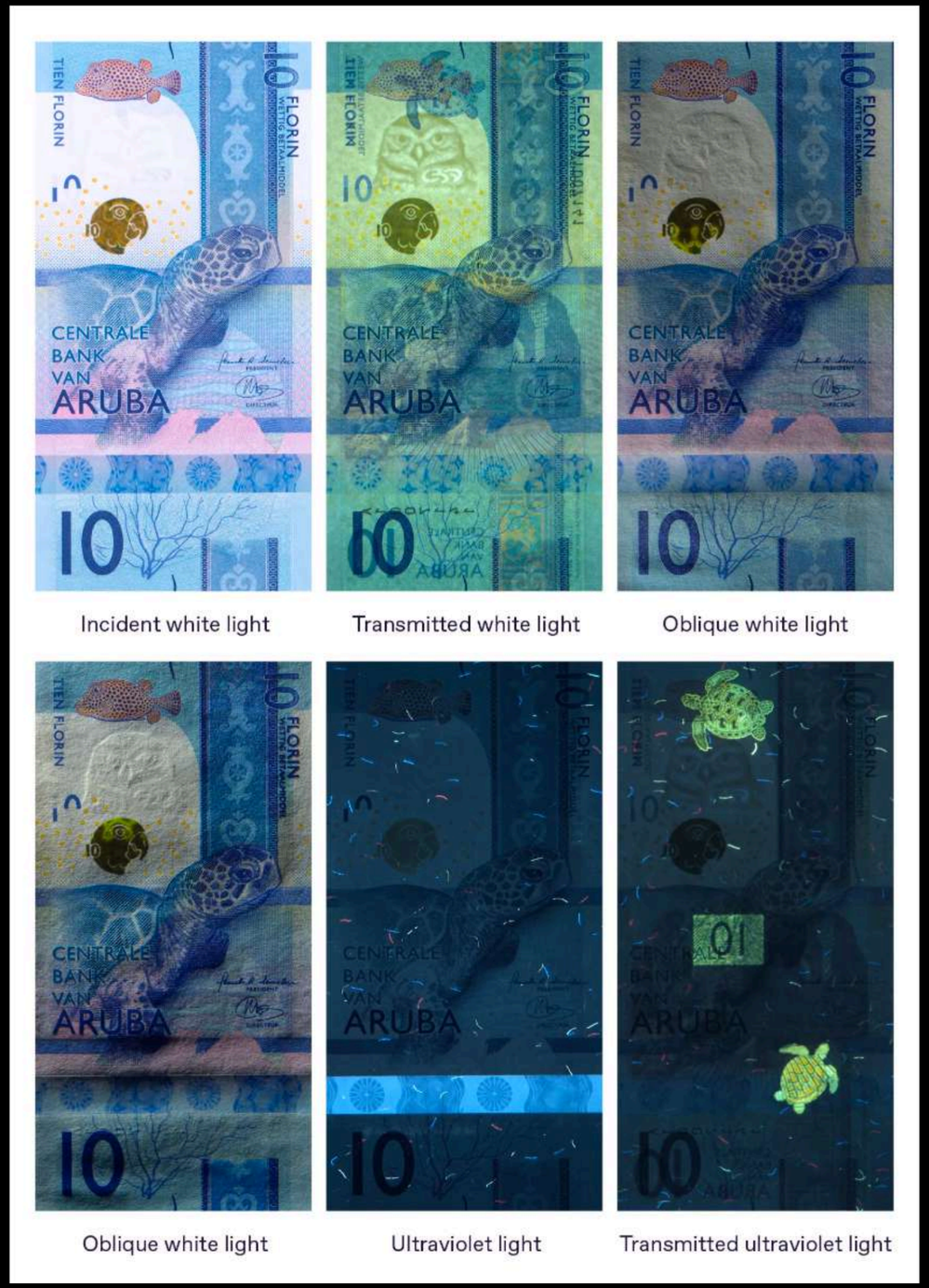
APPLICATION IN HOLOGRAM

- Nano-optic technology is emerging as a modern alternative to traditional holography in high-security applications, redefining the concept of Optically Variable Devices (OVDs).
- The technology marks a shift from conventional diffractive holograms to precise nanoscale control of light, using 3D microstructures of about 305 nm, it can accurately control the amplitude, phase, and polarization of light.
- This enables advanced visual effects such as 3D depth, motion, and colour shifting, based on metamaterial principles. Because the designs rely on high-precision fabrication and proprietary algorithms, they are extremely difficult to copy.
- As a result, nano-optic systems provide multi-level security, allowing quick public verification while also supporting detailed authentication with specialized instruments.

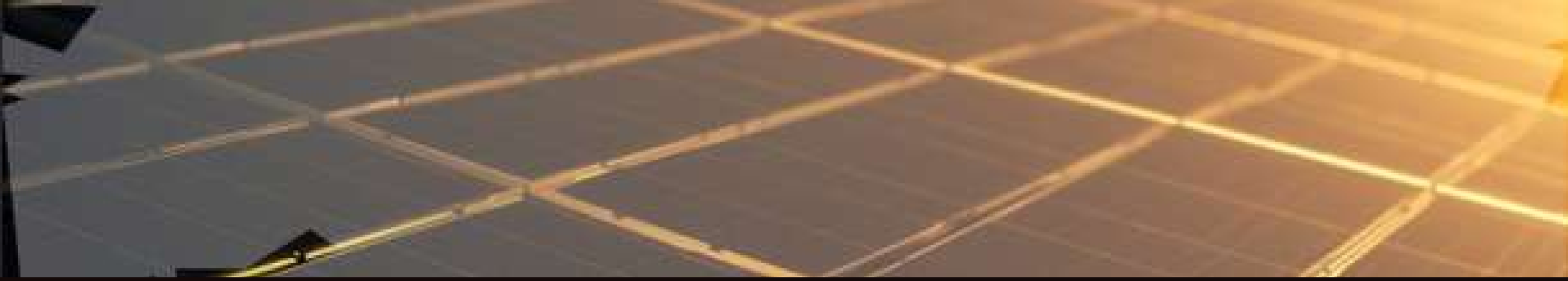
https://www.nanotec.or.th/en/butterfly-wings-behind-anti- counterfeiting-nanotechnology/
https://www.holostik.com/the-use-of-nano-optical-images-in- security-holograms/
https://www.kinegram.com/knowledge-base/kinegram- technology/kinegram-vs-hologram






TIEN FLORIN
10 FLORIN
CENTRALE BANK VAN ARUBA
10
Incident white light
Transmitted white light
Oblique white light
Oblique white light
Ultraviolet light
Transmitted ultraviolet light

SFU



APPLICATION IN SOLAR PANELS

Scientists use these principles to improve photovoltaic efficiency:

Solar panels are engineered with nano-patterned surfaces similar to butterfly wings.
These reduce:

- reflection losses
- increase absorption.

Mimicking the butterfly structure helps trap more sunlight inside the panel.
This is especially useful for:

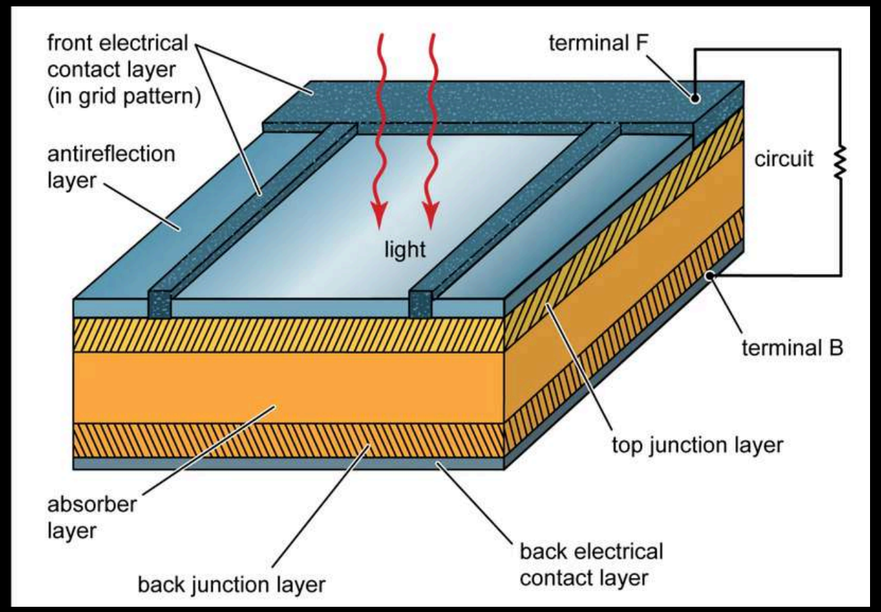
- Thin-film solar cells
- Low-light conditions

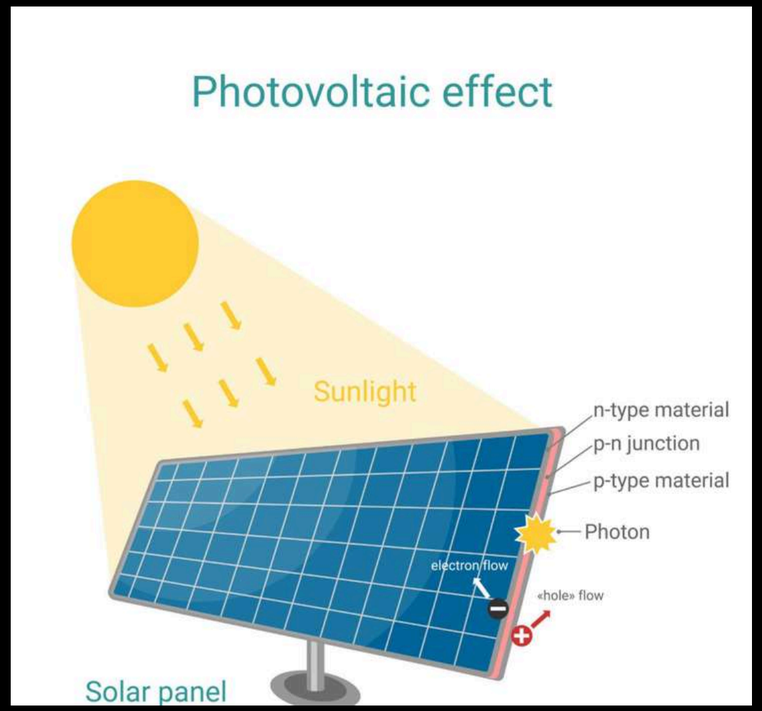
Researchers have created photonic crystal structures to control how light moves through solar cells.

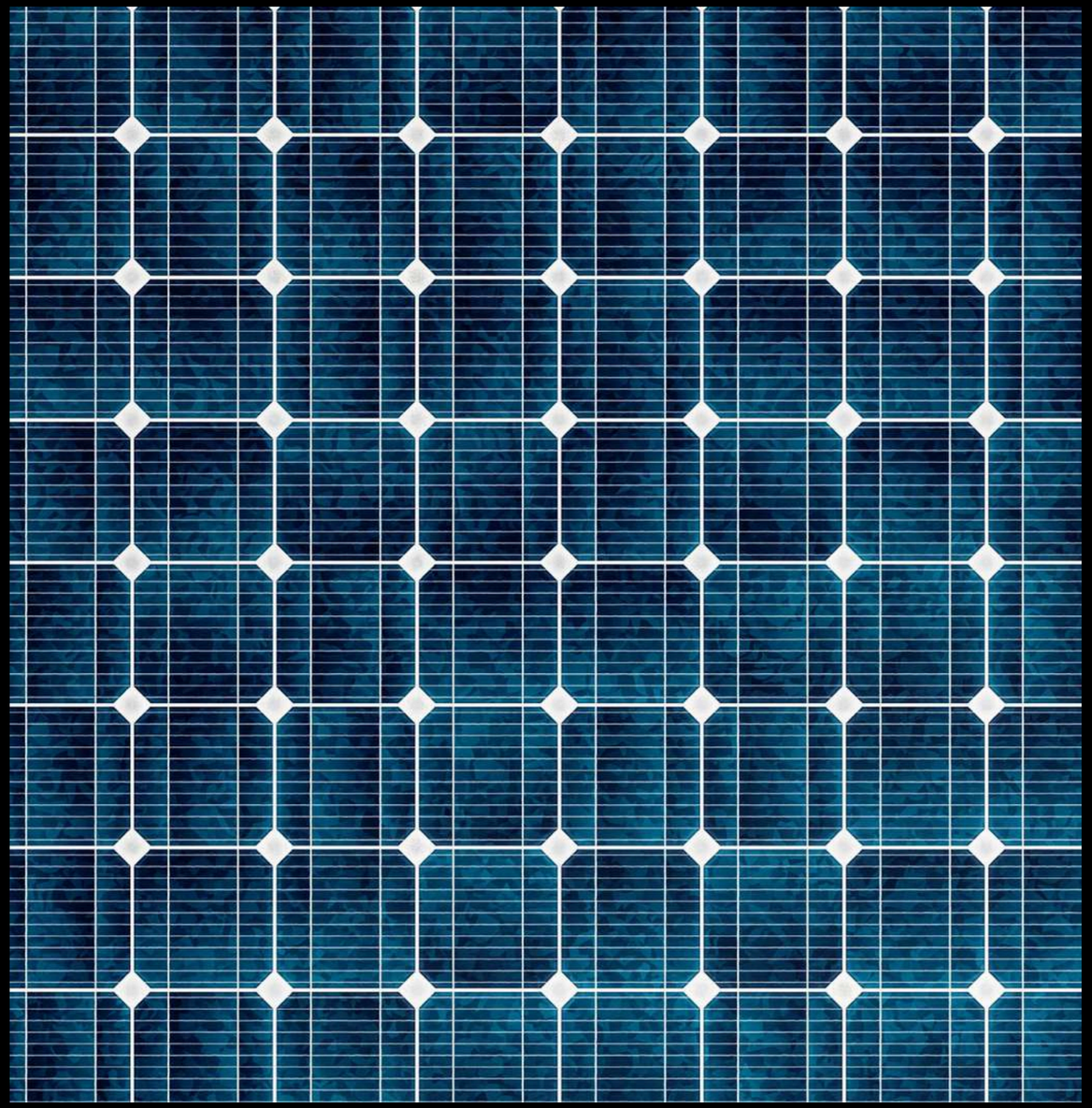
Real-World Impact:
Increased efficiency without significantly increasing material cost

- Better performance at different angles of sunlight
- Potential for thinner, lighter solar panels

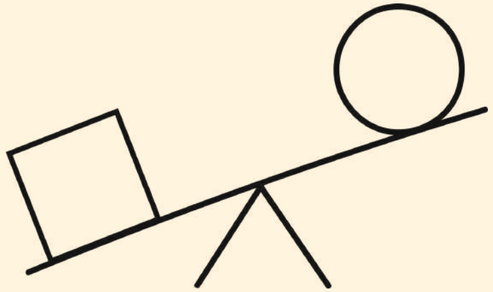
https://www.nature.com/articles/s41598-018-35082-3


front electrical
contact layer
(in grid pattern)
antireflection
layer
terminal F
circuit
light
terminal B
top junction layer
absorber
layer
back junction layer
back electrical
contact layer


Photovoltaic effect
Sunlight
n-type material
p-n junction
p-type material
Photon
electron flow
«hole» flow
Solar panel



PROS	CONS
I.)Unique optical effects that are difficult to replicate, improving security.	I.)High initial development and manufacturing cost.
II.)Supports authentication with visible and machine-readable features.	II.)Needs advanced fabrication techniques.
III.)Lightweight and durable, suitable for many materials. Eco-friendly, safer than chemical inks.	III.)Difficult to integrate into existing production lines.
IV.)Dynamic color effects that increase visual appeal and engagement.	IV.)Some features require special tools for verification.



https://en.wikipedia.org/wiki/Structural_coloration https://pmc.ncbi.nlm.nih.gov/articles/PMC8288439/ https://www.hoekmine.com/structural-colour/
https://www.researchgate.net/publication/232779014_Morpho_butterfly_wing_scales_demonstrate_highly_selective_vapour_response https://asknature.org/innovation/anti-counterfeiting-technology-inspired-by-the-morpho-butterfly/bit_of_body_text

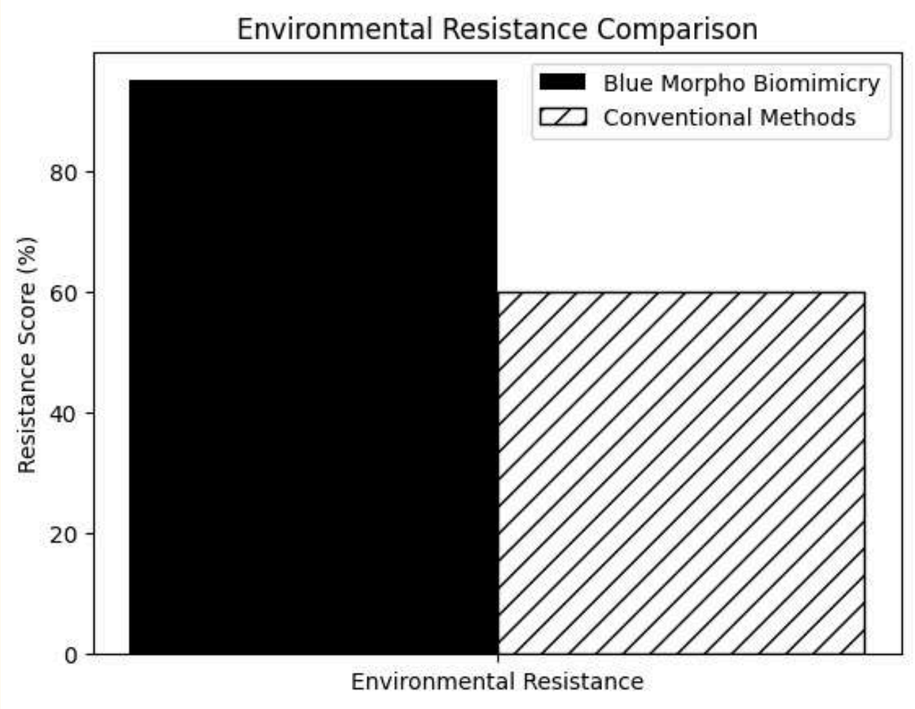
STATASTICAL DATA

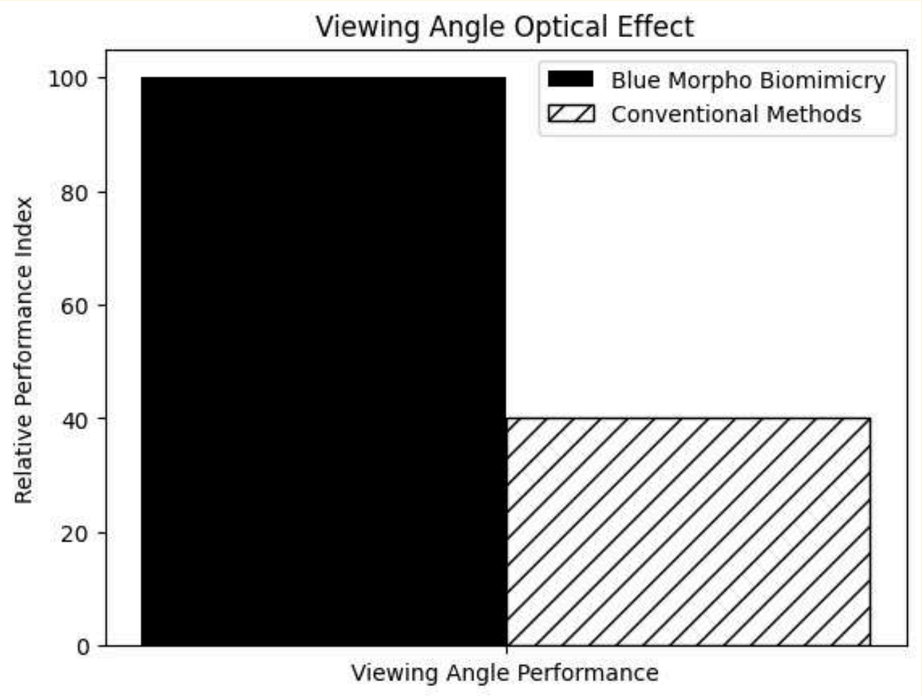
Performance Comparison: Morpho-Inspired Nano-Optics vs Conventional Optical Security Methods

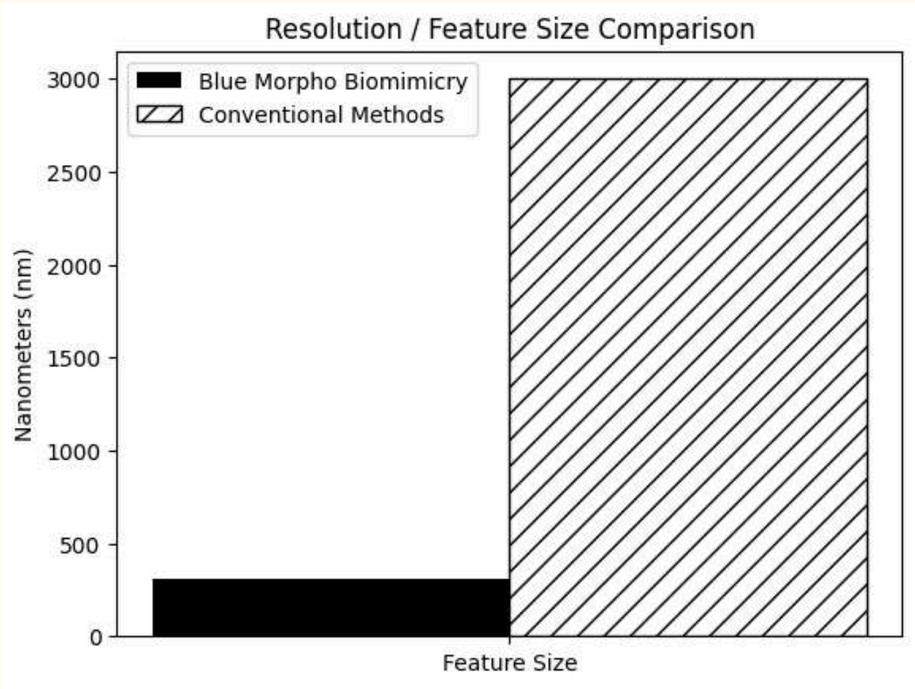
Parameter	Blue Morpho Biomimicry (Nano-Optic Structural Color)	Conventional Pigment / Holographic Methods
Durability (%)	95—100% stability	60—75% stability
Manufacturing Cost	$500—$1200 per m²	$50—$150 per m²
Security Level (Copy Resistance)	~99.9% physically unclonable nano-patterns	70–80% reproducible with lithography
Feature Size / Resolution	<305 nm (nano-scale precision)	1–5 µm (micro-scale precision)
Viewing Angle Effect	Strong 3D iridescence and color shift	Limited 2D color shift
Environmental Resistance	High resistance to UV, moisture, chemicals	Moderate resistance

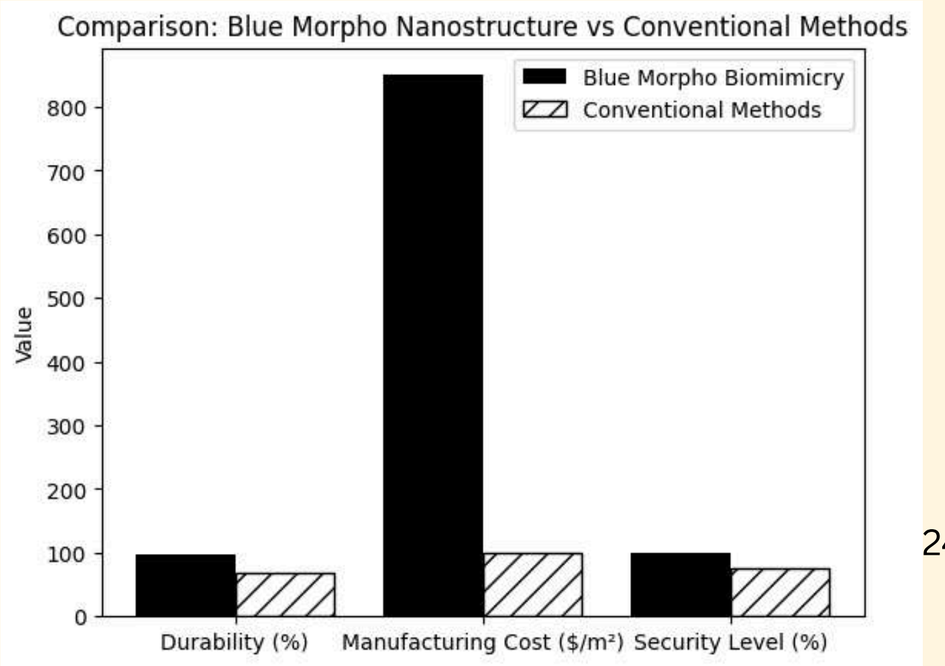
Performance Comparasion: Blue Morpho Biomimicry Nano-Optstructural Color) vs Conventional Pigment / Holographic Methods

https://www.nationalgeographic.com/culture/article/the-butterfly-effect-how-blue-morpho-wings-could-stop-counterfeiting
https://pubs.rsc.org/en/content/articlehtml/20I7/x0/d4na008I4f
https://www.sciencedaily.com/releases/20I0/05/I00530I44025.htm
https://textilevaluechain.in/in-depth-analysis/articles/biomimetics-structural-coloration-seen-in-nature


Environmental Resistance Comparison
Blue Morpho Biomimicry
Conventional Methods
Resistance Score (%)
0
20
40
60
80
Environmental Resistance


Viewing Angle Optical Effect
Blue Morpho Biomimicry
Conventional Methods
Relative Performance Index
0
20
40
60
80
100
Viewing Angle Performance


Resolution / Feature Size Comparison
Blue Morpho Biomimicry
Conventional Methods
Nanometers (nm)
0
500
1000
1500
2000
2500
3000
Feature Size


Comparison: Blue Morpho Nanostructure vs Conventional Methods
Blue Morpho Biomimicry
Conventional Methods
Value
0
100
200
300
400
500
600
700
800
Durability (%)
Manufacturing Cost ($/m²)
Security Level (%)

RESEARCH TIMELINE

1600-1800
(Natural observation of butterfly coloration)

Early 1900's
(Discovery of structural color)

1950-1970
(Electron microscopy reveals nanostructure)

1980-1990
(Photonic Crystal Theory)

2000-2010
(Biomimicry and artificial replication)

2010-.....
(Security printing application)

2015-.....
(Expansion into Solar cells and sensor)

https://en.wikipedia.org/wiki/Structural_coloration https://pmc.ncbi.nlm.nih.gov/articles/PMC8288439/ https://www.hoekmine.com/structural-colour/
https://www.researchgate.net/publication/232779014_Morpho_butterfly_wing_scales_demonstrate_highly_selective_vapour_response https://asknature.org/innovation/anti-counterfeiting-technology-inspired-by-the-morpho-butterfly/bit of body text

CONCLUSION

Butterfly wing-inspired nanostructures represent a significant advancement in security technology. Compared to traditional holograms, they provide better security, durability, and environmental sustainability. Many GIO central banks are already exploring this technology for future currency designs.

These nanostructures create visually striking patterns that are extremely difficult to replicate, making counterfeiting much harder. Since current counterfeiting rates range from about I7-32 fake notes per million banknotes, adopting this approach could greatly reduce fraud and improve the overall protection of modern banknotes.

THANK YOU SO MUCH!

END OF PRESENTATION